MIL-E-1/610E
18 March 1976
SUPERSEDING
MIL-E-1/610D
7 March 1969

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, THYRATRON

TYPES 3C23 AND 323B

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, gas and mercury

Mounting position: Vertical, base down

Weight: 3 ounces (85 grams) nominal

Type:		3C23		323B	
Base:		A4-10 (EIA)		A5-11 (EIA)	
Cap:		C1-5 (EIA)		C1-1 (EIA)	
Envelope:		ST-16 (EIA)		ST-16 (EIA)	
		Inches	(mm)	Inches	(mm)
Height:	Minimum:	5.625	142.88	6.000	152.40
	Maximum:	6.125	155.58	6.625	168.28
Diameter:	Maximum:	2.063	52.40	2.063	52.40

Base connections:	Pin No.	Element	Pin No.	Element
	1	f	1 and 2	f
	2	nc	3	g
	3	g	4 and 5	f
	4	f	cap	a
	cap	a		

ABSOLUTE RATINGS:

Parameter:	Ef	Ebb	epx	epy	ib	Ib	ig	THg	tk	Rg	Rp	F
Unit:	Vac	Vdc	v	v	a	Adc	ma	°C	sec	Ohms	Ohms	Hz
Maximum:	2.625	---	1,250	1,250	6.0	1.5	50	+80	---	---	---	---
Minimum:	2.375	---	---	---	---	---	---	-55	15	---	---	---
TEST CONDITIONS:	2.5	500	---	---	---	---	---	---	15	1,000	5,000	60

GENERAL:

Qualification - Required

(E) denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	AQL (PERCENT DEFECTIVE)	INSPECTION LEVEL OR CODE	SYMBOL	LIMITS		UNIT
							MIN	MAX	
	Quality conformance inspection, part 1								
3241	Heater current	-		0.65	II	If	6.25	7.75	Aac
3201	Critical grid voltage for conduction (1)	-	THg = 40° to 70°C; Rp = 350 to 5,000 ohms; vary Ecc	0.65	II	Ecc	-2.0	-5.5	Vdc
3201	Critical anode voltage for conduction	-	THg = 40° to 70°C; Rp = 25 to 5,000 ohms; Ecc = 0; vary Ebb	0.65	II	Ebb	10.0	35.0	Vdc
(E) 1231	Peak emission by voltage drop	1	THg = 40°C (nominal); Ib = 8.0 a	0.65	II	etd	---	12.0	v
(E) 3206	Operation (2)	1	THg = 40°C (nominal); epy = epx = 1,500 v; Rg = 100,000 ohms; vary Ecc	0.65	II	Ecc	-4.0	-12.0	Vdc
3211	Cold-gas conduction	-	Rp = 1.5 Meg; Ib = 2 µAdc (min)	0.65	II	Epp	2,000	5,000	Vac
	Quality conformance inspection, part 2								
1031	Low-frequency vibration	-	No voltages applied	---	---	---	---	---	---
3201	Critical grid voltage for conduction (2)	2	THg = 40° to 70°C; Ebb = 100 Vdc; vary Ecc	---	---	ΔEcc	---	1.9	Vdc
3206	Operation (1)	-	THg = 78° to 82°C; epy = epx = 1,500 v; Rg = 100,000 ohms; vary Ecc	---	---	Ecc	-4.0	-12.0	Vdc
3216	Grid current	3	Ep = 220 Vac 10%; Rp/Ib = 1.5 Adc; Ef = 2.75 Vac; Rg = 1.0 Meg (min)	---	---	Ic	---	5.0	µAdc
	Quality conformance inspection, part 3								
---	Life test	-	Group C; Epp = 220 Vac; Rp/Ib = 1.5 Adc; Ecc = 0; Rg = 10,000 ohms; t = 500 hours	---	---	---	---	---	---
---	Life-test end points:								
1231	Peak emission by voltage drop	-		---	---	etd	---	20	v
3216	Grid current	-		---	---	Ic	---	5	µAdc
3206	Operation (1)	-		---	---	Ecc	-4.0	-13.0	Vdc
3206	Operation (3)	4,5	epy = epx = 1,500 v; Rg = 100,000 ohms; THg = -55°C; vary Ecc	---	---	Ecc	-4.0	-12.0	Vdc

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NOTES:

(E) 1. The ambient temperature shall be between 20° and 40°C and artificial cooling shall not be used.

2. The difference between the critical grid voltage, Ecc at Ebb = 100 Vdc and that at Ebb = 500 Vdc, shall not exceed the limit specified.

3. Tube shall be operated 5 minutes immediately before the test.

(E) 4. This test shall be performed during the initial production and once each succeeding 12-calendar months in which there is production. A regular double sampling plan shall be used, with the first sample of three tubes with an acceptance number of zero, and a second sample of three tubes with a combined acceptance number of one. In the event of failure, the test will be made as part of quality conformance inspection, part 2, code level D, with an AQL of 6.5. The regular "12-calendar month" double sampling plan shall be reinstated after three consecutive samples have been accepted.

(E) 5. Allow tube to stabilize nonoperating for a minimum of 5 minutes at the specified cold temperature. Apply Ef 15 seconds prior to application of anode supply voltage. Tube shall conduct upon application of anode voltage at specified minimum Ecc and conduction shall stop before the specified maximum Ecc is reached.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 80
DSA - ES

User activities:
Army - WC, MU
Navy - AS, OS, MC, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
DSA - ES

(Project 5960-3078)

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